Chapter 6.1 — Formula Sheet

Dynamic Programming on Random Variables

Companion quick-reference to your Notebook 6 syllabus, your *20 Worked Examples* (Ex 1–20) and the *Stage-1 Hard Drill* (Q1–Q10)

A. Backward Induction

Master Recursion [Ex 1–2; Q1]

Up to k draws; V_j = value at the start of draw j:

$$V_k = \mathbb{E}[X], \qquad V_j = \mathbb{E}\big[\max(X, V_{j+1})\big]$$

Accept X iff $X > V_{j+1}$ (the reservation value). Always start from the last decision and work back.

Fast $\mathbb{E}[\max(X,t)]$ [Ex 3–4; Q1, Q6]

Faces below t are replaced by t. Fair d-sided die, $m = \lfloor t \rfloor$:

$$\mathbb{E}[\max(X,t)] = \frac{m\,t + \left[\frac{d(d+1)}{2} - \frac{m(m+1)}{2}\right]}{d}$$

d6: 3.5, $\frac{17}{4}$, $\frac{14}{3}$, $\frac{89}{18}$, $\frac{277}{54}$ for $k = 1..5$ (gains $0.75, 0.42, 0.28, 0.19$: diminishing). d20: 10.5, 13, 14.4.

Running the Optimal Rule [Ex 5, 10; Q7]

$$\mathbb{P}(\text{stop at } j) = \mathbb{P}(\text{reach } j)\,\mathbb{P}(X > V_{j+1})$$

$$\mathbb{E}[\#\text{rolls}] = 1 + \sum_{j \geq 2} \mathbb{P}(\text{reach } j)$$

d6, $k = 3$: $\mathbb{E}[\#] = 2$, final payoff $\mathbb{P}(6) = \mathbb{P}(5) = \frac{1}{3}$. Evaluate any *fixed* rule by the same backward recursion [Ex 9]: the cost of a wrong threshold is $V_{\text{opt}} - V_{\text{rule}}$.

Put the Objective at the Leaves [Q2, Q3]

Optimising $g(X)$: run the DP on g, never on X:

$$V_k = \mathbb{E}[g(X)], \quad V_j = \mathbb{E}[\max(g(X), V_{j+1})]$$

$\mathbb{E}[g(X^\star)] \neq g(\mathbb{E}[X^\star])$ (gap = Var for $g = x^2$). Win probability vs a single die: $g(x) = \frac{2x-1}{12}$ is linear, so the plain rule is optimal.

B. Other Distributions

General Identity [Ex 6, 8]

$$\mathbb{E}[\max(X,t)] = t + \mathbb{E}[(X-t)^+]$$

Skewed die: the net is the mean of the *whole* die, not its middle face ($\{2,3,5,8,13,21\}$: net $\frac{26}{3}$ rejects an 8).

Uniform$(0,1)$ [Ex 7, 18; Q4]

$$\mathbb{E}[\max(U,t)] = \frac{1+t^2}{2}, \qquad \mathbb{E}[\min(U,t)] = t - \frac{t^2}{2}$$

Sell side $V \mapsto \frac{1+V^2}{2}$: $\frac{1}{2}, \frac{5}{8}, \frac{89}{128}, \ldots$ Buying (minimise) is the mirror: $V^{\text{buy}} = 1 - V^{\text{sell}}$ ($\frac{39}{128}$ for 3 quotes). Shortfall $1 - V_n \approx \frac{2}{n+c}$: 16 draws to reach 0.9.

Exponential & Normal [Ex 8; Q5]

$$X \sim \text{Exp}(1): \quad \mathbb{E}[\max(X,t)] = t + e^{-t}$$

$$Z \sim \mathcal{N}(0,1): \quad \mathbb{E}[\max(Z,t)] = t\Phi(t) + \varphi(t)$$

Exp: 1, 1.368, 1.623 for 1, 2, 3 draws. Normal: 0, $\frac{1}{\sqrt{2\pi}} = 0.399$, 0.630.

Gambler vs Prophet [Q9; Ex 3, 4, 7]

Prophet sees all draws: $\mathbb{E}[\max] = \sum_k \mathbb{P}(\max \geq k)$. Two d6: $\frac{161}{36}$ vs stopper $\frac{17}{4}$ (ratio 0.950). Uniform, 3 draws: $\frac{3}{4}$ vs $\frac{89}{128}$ (0.927). iid worst-case ratio ≈ 0.745. Bigger die beats more tries: 8 rolls of d12 $\approx$ one d20 roll [Q6].

C. Stopping With Costs

Finite Horizon With a Re-roll Fee [Ex 13, 19, 20]

First draw free, each re-draw costs c:

$$V_j = \mathbb{E}\big[\max(X,\, V_{j+1} - c)\big]$$

d6, $c = \frac{1}{2}$, $k = 3$: $\frac{17}{4}$ (= two free rolls). $c = \frac{1}{4}$, $k = 4$: $\frac{37}{8}$.

Early-Exit Fee [Ex 14]

Taking the roll before the last costs e:

$$V_j = \mathbb{E}\big[\max(X - e,\, V_{j+1})\big] \ (j < k), \qquad V_k = \mathbb{E}[X]$$

d6, $e = \frac{1}{2}$, $k = 3$: $\frac{13}{3}$.

Unlimited Draws, Every Draw Costs c [Ex 11–12; Q10]

Continuation value $C = -c + \mathbb{E}[\max(X, C)]$, equivalently

$$\boxed{\mathbb{E}\big[(X-C)^+\big] = c}$$

Stop iff $X > C$; $\mathbb{E}[\#] = \frac{1}{\mathbb{P}(X>C)}$; net $= \mathbb{E}[X \mid X > C] - c\,\mathbb{E}[\#] = C$. d6: $c = \frac{1}{4}, \frac{1}{2}, 1 \to C = \frac{19}{4}, 4, 3$. If $c \geq \mathbb{E}[X] - \min X$, take the first offer.

Closed Forms for C [Ex 15–16]

$$U(0,1): \ C = 1 - \sqrt{2c} \qquad X \sim \text{Exp}(1): \ C = \ln \tfrac{1}{c}$$

Uniform $c = 0.02 \to 0.8$ (5 draws); Exp $c = 0.1 \to 2.303$ (10 draws). Standard grows only like $\ln(1/c)$ for Exp.

Discount / Game-Ending Risk [Ex 17; Q7]

Passing is worth δV (δ = survival or discount factor):

$$V = \mathbb{E}[\max(X, \delta V)]$$

Guess the accept set, solve the linear equation, verify consistency. d6, $\delta = 0.9$: $V = \frac{11}{6-4\delta} = \frac{55}{12}$. Casino closes w.p. $\frac{1}{4}$ ($\delta = \frac{3}{4}$): $V = 4$, $\mathbb{P}(0) = \frac{1}{9}$.

D. Capstones

One-Step Look-Ahead [Q8]

Bust game (face 1 loses the bank b; faces 2–6 add):

$$\text{gain} = \frac{5b+20}{6} - b = \frac{20-b}{6}$$

Gain positive iff $b < 20$ and shrinks as b grows, so stopping is a closed set: roll while $b < 20$. $V(0) \approx 8.14$. Rule: compare expected gain with expected loss.

Fee $\leftrightarrow$ Cap Exchange Rate [Q10]

Break-even fee where unlimited paid rolls equal k free rolls: set $C = V_1^{(k)}$ in $\mathbb{E}[(X-C)^+] = c$. d6, $k = 3$ ($C = \frac{14}{3}$):

$$c^\star = \frac{(5 - \frac{14}{3}) + (6 - \frac{14}{3})}{6} = \frac{5}{18}$$

Cap on tries and fee per try both ration patience.

Reminder: start at the last decision; the value of passing is the net you compare against (V_{j+1}, $V_{j+1} - c$, δV or C). Put the quantity you actually optimise at the leaves, guess-and-verify stationary rules, and check every answer by simulating the rule.